

Protecting Salem's Ecosystem

Predictive Modeling of Tree Canopies and Stream Temperatures

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Abstract

Rising stream temperatures are threatening Salem's ecosystem due to many environmental factors. The loss in tree canopy coverage and increasing temperatures risk pushing native species to cooler bodies of water and away from Salem's waterways. Here, we aimed to model the change in Salem's Mill Creek temperature from 2015 to 2025 using historical stream temperature, weather, and canopy coverage data using linear regression models. One model stood out among the rest, with an adjusted R-squared value of 0.9442. This model combines high-level interactions within temperature and time to predict stream temperatures. Within the model, with the available data, tree canopy does not have a significant impact on stream temperature. The main driver of the model is time. This means that, as it stands, more work should be done in protecting stream temperature against future rising temperatures, regardless of tree canopy coverage. A defining challenge in this project included a lack of reliable data; thus, it is also recommended that the City of Salem allocate more resources to collecting data that would shed more light on factors impacting stream temperature.



Introduction & Motivation

Problem & question:

The City of Salem is concerned about increasing stream temperatures and the impact that this will have on the environment. Stream temperature rise can impact indigenous communities who rely on Salmon populations, citizens who utilize streams for cooling or drinking, and anyone else who uses water in Salem. Additionally, rising stream temperatures threaten local species and spur harmful algal blooms, posing serious risks to Salem's population.

- Can historic stream temperature data be used to predict stream temperature with an R-squared value of at least 90%?
 - Will adding historic weather data for Salem, Oregon improve the R-squared value from the original model?
 - Will adding historic tree canopy data for Salem, Oregon improve the R-squared value from the original model?
- Can a model be used to predict Salem's stream temperature 3 years into the future based on different levels of anticipated climate change impact on the weather data?



Background & Related Work

Bodies of water around the world are beginning to heat up at faster rates than air heat waves [1]. These water heat waves—defined by at least five consecutive days above the seasonal average—result in lower dissolved oxygen content, nutrient cycling, and pH buffering [3]. All of this means that ecosystems and native species adapted to live in colder waters face serious threats, while algae has room to become prolific.

One way to combat high stream temperatures is to support riparian zones, the dense plant areas along rivers [4]. These areas often sport trees and other foliage that shade and cool rivers and streams, effectively lowering the water temperature. However, these areas are at risk for several reasons.

This recent expansion of threats to river temperatures has caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan [2]. No research currently exists on stream temperatures, weather, and tree canopies in Salem, to our knowledge. This research will support government and private organizations in the prevention of the loss or disruption of riparian zones in Salem.

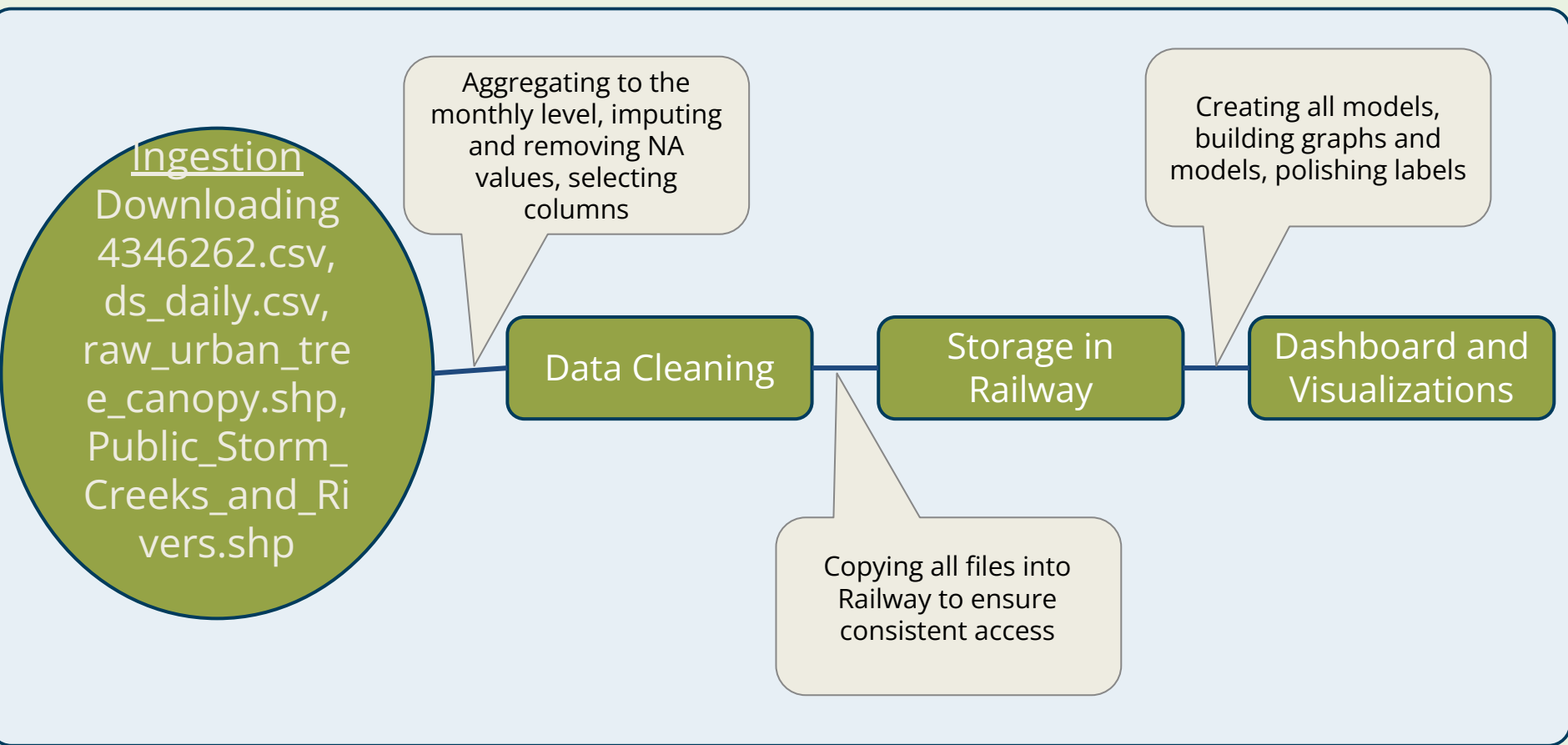


Data, Engineering & Ethics

Sources & scale: The data is sourced from NOAA, The City of Salem, Tree Plotter Canopy, and CastawayWU. These were all available with permission from the data publisher, and could be cleaning downloaded as a flat CSV file.

Ethics & responsible use: Our data contains no PII, so we are not concerned about accidentally sharing people's personal information. All of our data is publicly available, so we are not concerned about consent or redaction. Our main ethical concern with this project is creating predictions with the limited years of data that we have access to. This risks missing overall trends that are seen across years. This also limits our ability to make predictions into the future, rather we must focus on emphasising the scope of stream temperature change in Salem between years.

Pipeline:



Methods & Evaluation

Approach:

To answer the research questions, linear regression models with high-level interactions were created in R iteratively, with different data included for each model. The linear regression models predicting stream temperature were done in the following order:

- Using 2015-2025 stream temperature data only
- Using 2015-2025 stream temperature and weather data
- Using 2020, 2022, and 2024 stream temperature, weather, and tree canopy data

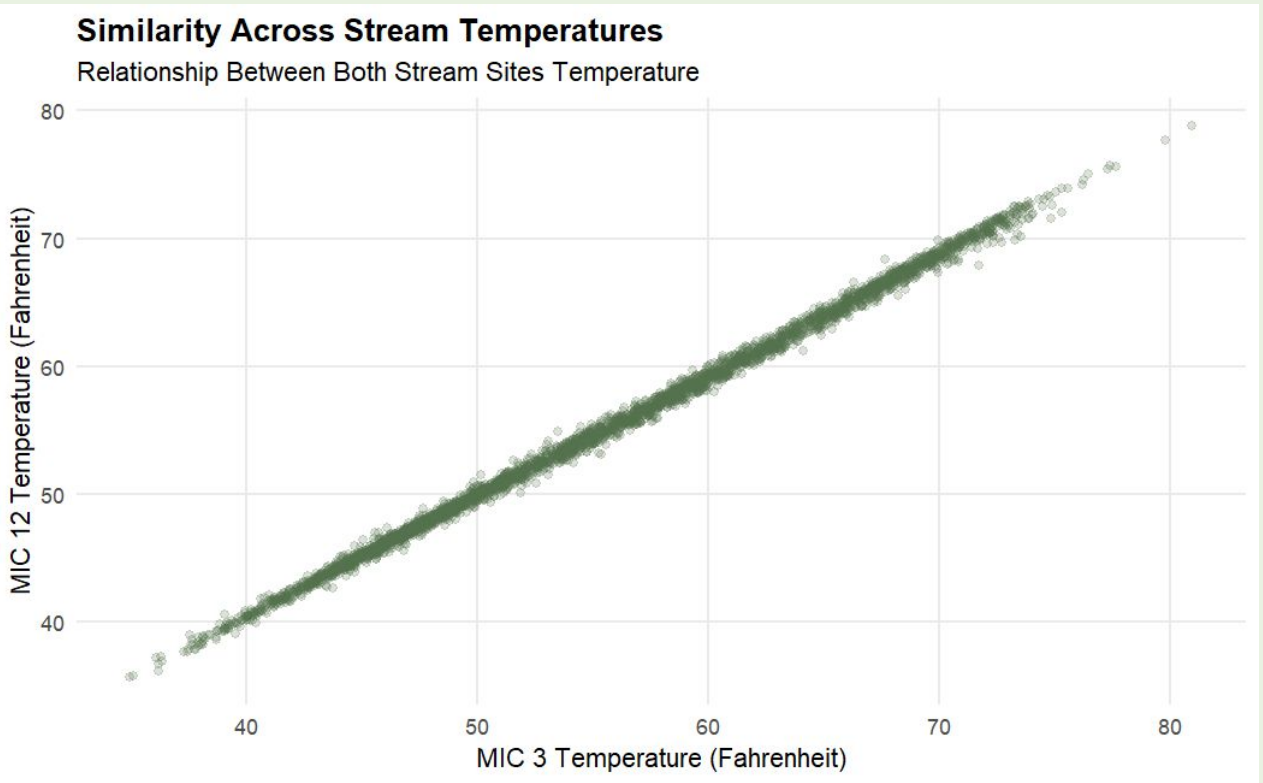
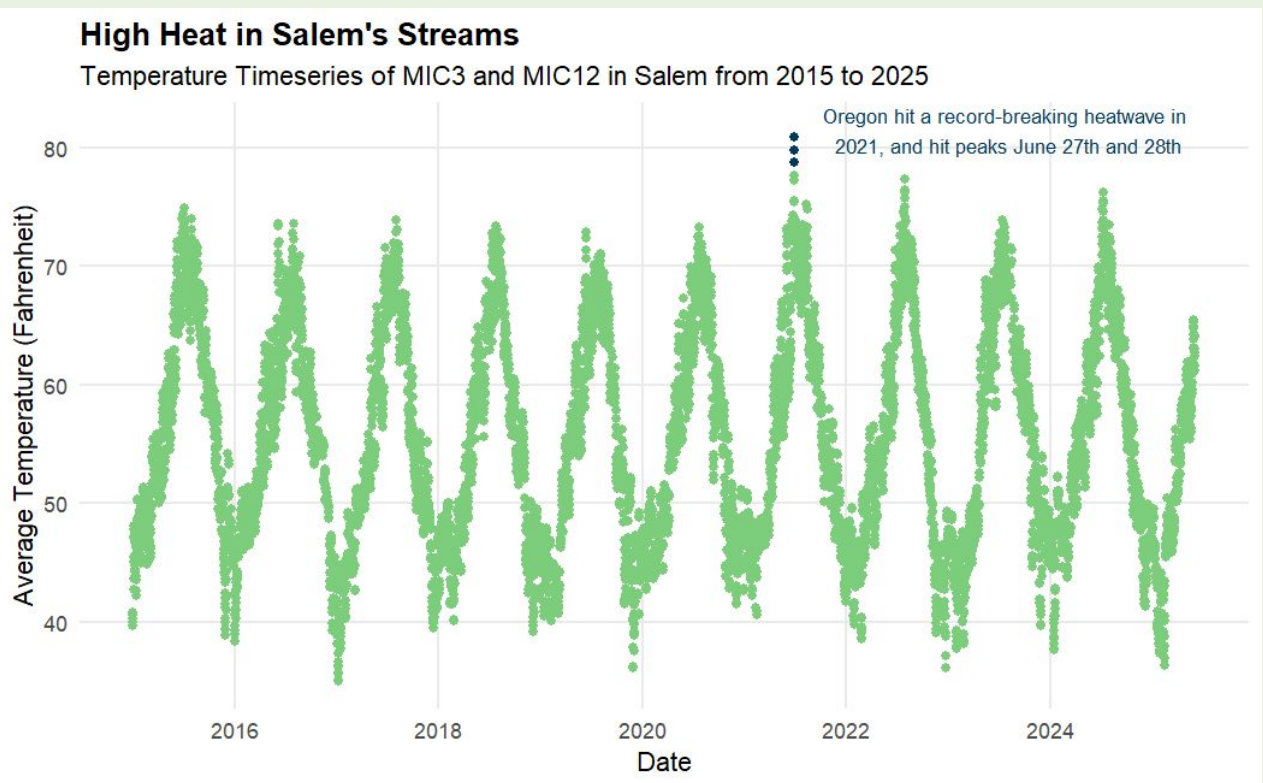
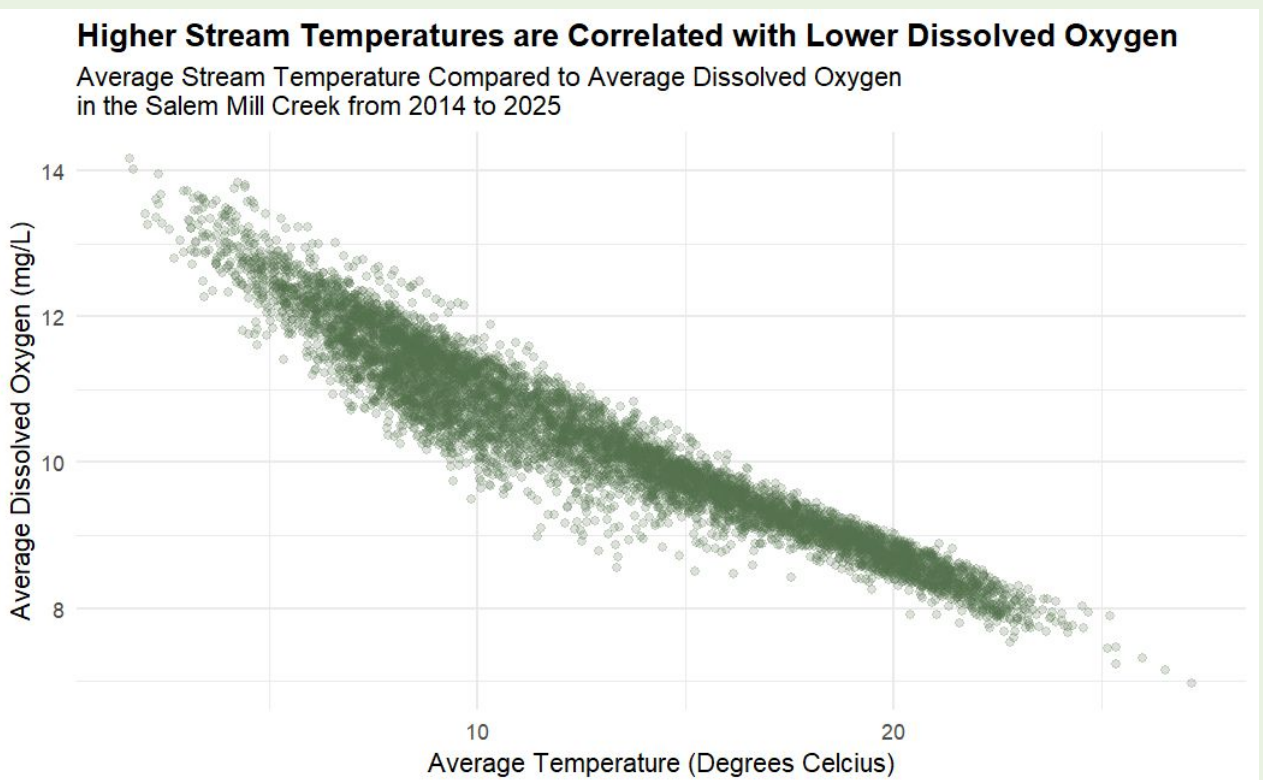
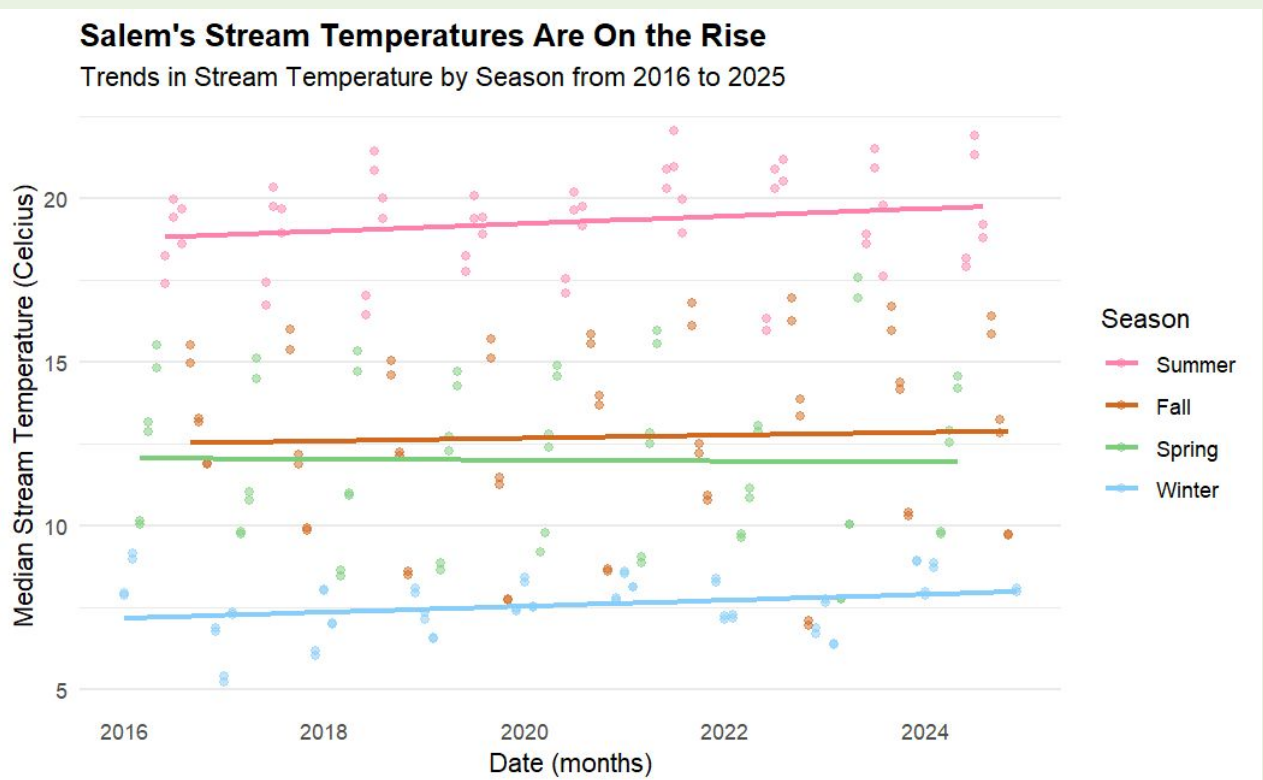
Validation:

Text



Results & Visualizations

After several model iterations of linear regression, we landed on one that gives the high level interactions between precipitation and air temperature, as well as day of the month and month, in addition to finding the overall relationship between the average temperature and location, year, month, precipitation, air temperature, and day of the month. This model produces an adjusted R-squared value of 0.9442.



Discussion, Limitations & Conclusions

Discussion:

Within our model, with the available data, tree canopy does not have a significant impact on stream temperature. The main driver of the model is time. This means that as it stands, more work should be done in protecting stream temperature against future rising temperatures, regardless of tree canopy coverage.

Limitations & future work:

Our data is extremely limited due to only having 3 years of data for tree canopy data.

In the future, we can use the models we created as a base to create a new model with more years of data, or adding more variables such as insect infestations or human tree removal patterns. Our current best model does not incorporate tree information because it is not significant at the granularity available. As data on both tree canopy and Emerald Ash Borer prevalence becomes more widely available and well-documented, this model will likely benefit from the incorporation of that data.

Conclusions:

We can accurately predict stream temperatures within the year of the data, with or without canopy data. Due to our findings, we have two main calls to action:

- Protecting tree canopy along waterways in Salem is important for maintaining healthy stream temperature
- Further data collection would greatly impact the ability to model risks to stream temperature, particularly for tree canopy information.
 - A great way to get information on tree canopies in Salem would be LiDAR data for each year in the future.



Reproducibility & References

Repository: <https://github.com/Serenna4/Emerald-Ash-Borer-Capstone-Project/tree/main>
Key dependencies: R 4.5.2, Beekeeper SQL 5.9, Railway | Libraries: tidyverse, dplyr, MASS, car, forecast, astsa
References: [1] Bush, E., U.S. rivers are experiencing unprecedented and unexpectedly intense warming, NBC News, 2025.
[2] City of Salem, City of Salem Emerald Ash Borer Readiness and Response Plan, City of Salem, 2026.
[3] Penn State Extension, We are in Hot Water—Streams and Rivers are Heating Up, Pennsylvania State University, 2026
[4] University of Georgia, Stream Warming and Brook Trout in the Southeast, University of Georgia, 2026.

Acknowledgements

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